

Amelia Nair

Embedded Systems & IoT Firmware Engineer • 8 Years Experience • MERIQ Index: 75% (Strong Candidate)

■ amelia.nair54@example.com | ■ +1 (437) 732-7767 | ■ Singapore

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: PCB Debugging, C/C++, MQTT, I2C/SPI/UART, Python, Embedded Linux

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core PCB Debugging service, reducing latency by 41%.
- Led a squad of 5 engineers delivering scalable microservices processing 30M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 74%.

Associate Embedded Systems & IoT Firmware Engineer — *Tesla* (2018 - 2023)

- Designed and maintained C/C++ data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using MQTT saving \$30k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [PCB Debugging, C/C++, MQTT, I2C/SPI/UART]

Engineered end-to-end production solution leveraging PCB Debugging and C/C++ with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [MQTT, I2C/SPI/UART, Python, Embedded Linux]

Architected high-resilience system integrating MQTT with 99.9% uptime.

EDUCATION & CREDENTIALS

B.S. in Computer Science

IIIT Hyderabad — 2014 - 2018 (Score: 3.46 / 4.0)